

Probability - Homework 2

B13902022 LAI,YU-CHI

2.1-7**(a)**

Let the pmf here be $f(x)$, equivalent to the probability $P(X = x)$. ($x \in \{1, 2, 3, 4, 5, 6\}$ since the minimum value must be within the range)

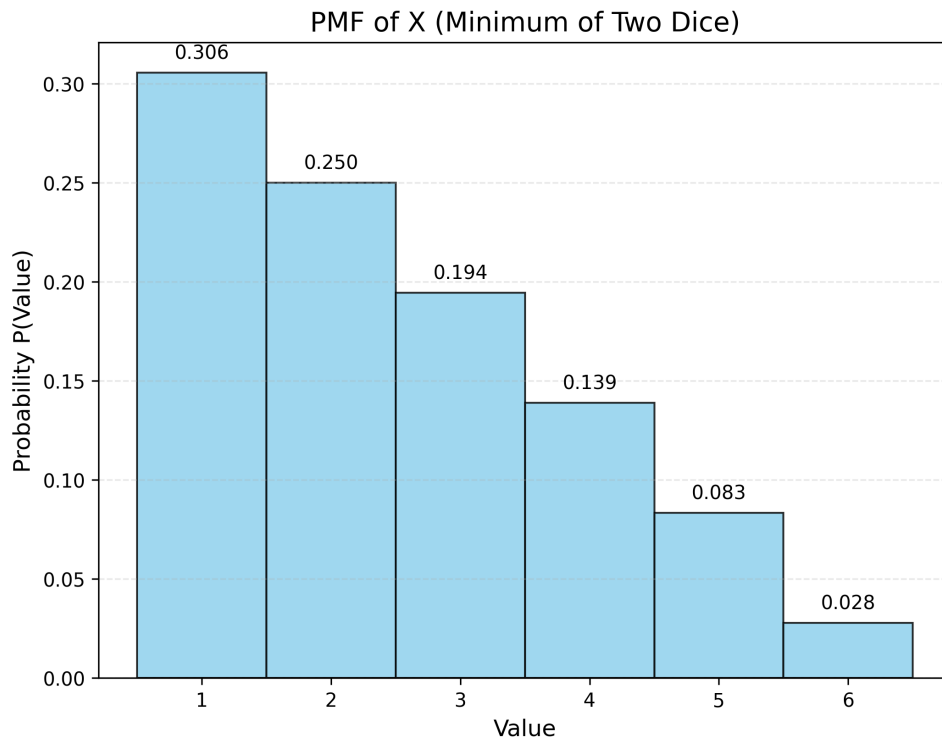
Let the minimum value of two outcomes be x , then we can choose one dice to roll that point, where the probability is $\frac{1}{6}$, and the other dice must roll out a point greater than or equal to x , which has the probability of $\frac{7-x}{6}$. Hence, the pmf $f(x) = \binom{2}{1} \cdot \frac{1}{6} \cdot \frac{7-x}{6} - \frac{1}{6} \cdot \frac{1}{6} = \frac{13-2x}{36}$. (The subtraction here is because we repeatedly count the condition that both dices roll out the same (minimum) point).

$$P(X = x) = f(x) = \frac{13 - 2x}{36}$$

(b)

Using the result in (a), we obtain:

$$\begin{aligned} f(1) &= \frac{11}{36} \\ f(2) &= \frac{9}{36} = \frac{1}{4} \\ f(3) &= \frac{7}{36} \\ f(4) &= \frac{5}{36} \\ f(5) &= \frac{3}{36} = \frac{1}{12} \\ f(6) &= \frac{1}{36} \end{aligned}$$

Figure 1: The histogram for pmf $f(x)$

(c)

Here (a, b) denotes that the first dice (here the term first and second just means the two dices are distinct) rolls out a and the second dice rolls out b .

If $y = 0$, there are 6 possibilities (two dices have the same point).

If $y = 1$, there are 10 possibilities $((1, 2), (2, 3), (3, 4), (4, 5), (5, 6), (2, 1), (3, 2), (4, 3), (5, 4), (6, 5))$.

If $y = 2$, there are 8 possibilities $((1, 3), (2, 4), (3, 5), (4, 6), (3, 1), (4, 2), (5, 3), (6, 4))$.

If $y = 3$, there are 6 possibilities $((1, 4), (2, 5), (3, 6), (4, 1), (5, 2), (6, 3))$.

If $y = 4$, there are 4 possibilities $((1, 5), (2, 6), (5, 1), (6, 2))$.

If $y = 5$, there are 2 possibilities $((1, 6), (6, 1))$.

The total number of possibilities is $6 \times 6 = 36$. Hence:

$$\begin{aligned}
 g(0) &= \frac{6}{36} = \frac{1}{6} \\
 g(1) &= \frac{10}{36} = \frac{5}{18} \\
 g(2) &= \frac{8}{36} = \frac{2}{9} \\
 g(3) &= \frac{6}{36} = \frac{1}{6} \\
 g(4) &= \frac{4}{36} = \frac{1}{9} \\
 g(5) &= \frac{2}{36} = \frac{1}{18}
 \end{aligned}$$

A more beautiful form of g maybe is ($y \in \{0, 1, 2, 3, 4, 5\}$):

$$g(y) = \begin{cases} 6 & \text{if } y = 0 \\ \frac{12-2y}{36} & \text{if } y \neq 0 \end{cases} \quad (1)$$

(d)

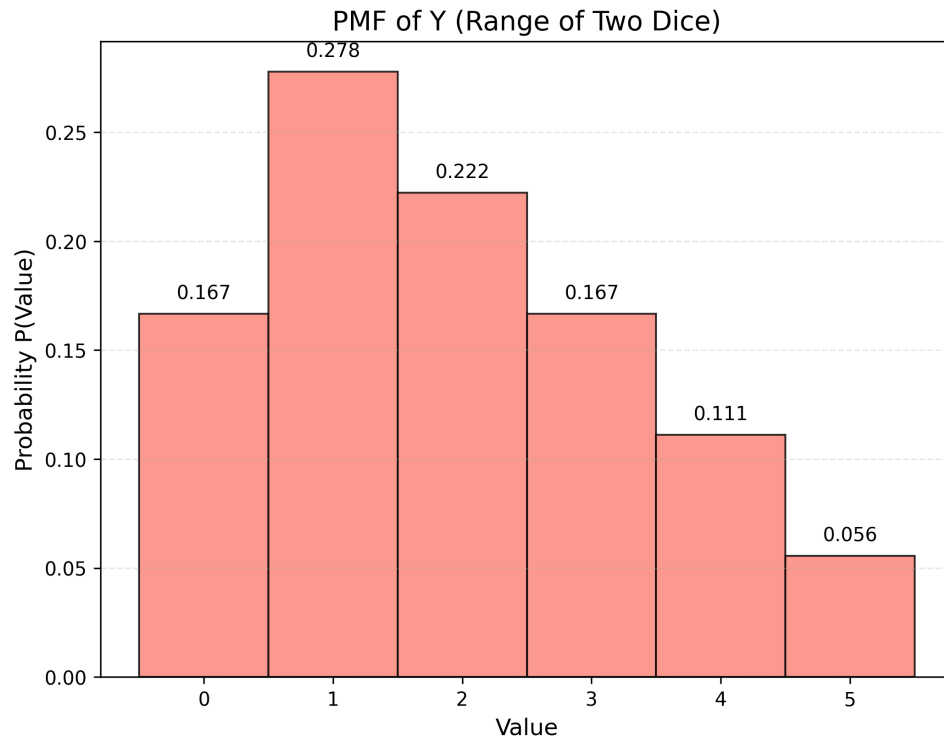


Figure 2: The histogram for pmf $g(y)$

2.2-10

By some simple calculations (Just count the number of possible outcomes), the probability of rolling out the sum in $\{2, 3, 4, 5, 6\}$ is $\frac{1+2+3+4+5}{36} = \frac{15}{36}$, and the probability of rolling out the sum in $\{8, 9, 10, 11, 12\}$ is $\frac{5+4+3+2+1}{36} = \frac{15}{36}$, the probability of rolling out 7 is $\frac{6}{36}$

We use the formula $\sum x_i P(X = x_i)$ to calculate the expected value, where x is the net gain or loss, X is the random variable for that. Hence, if you bet low, the expected value is $1 \cdot \frac{15}{36} + (-1)(1 - \frac{15}{36}) = \frac{-6}{36} \approx -0.1667$. If you bet high, the expected value is $1 \cdot \frac{15}{36} + (-1)(1 - \frac{15}{36}) = \frac{-6}{36} \approx -0.1667$. If you bet 7, the expected value is $4 \cdot \frac{6}{36} + (-1)(1 - \frac{6}{36}) = \frac{-6}{36} \approx -0.1667$.

The unit here is \$

2.2-11

The expected value of X (how much can a player earn in one round of game) is:

$$E[x] = 1 \cdot 0.49293 + (-1) \cdot 0.50707 = -0.01414$$

Thus, the answer is -\$0.01414.

2.3-16

(a)

Let the pmf be $f(x)$. If $X = 0, 1$, $P(X = 1) = f(1) = 0$, $P(X = 0) = f(0) = 0$. For $X > 1$, we can observe that there are only two possibilities, one is the last two flip being both head, the other one is the last two flip being both tail. The two possibilities yield exactly two distinct permutations (e.g. for $X = 4$, the only results are (h, t, h, h) and (t, h, t, t) , where the tuple $(\dots)$ represents the result of each round, h means head, while t means tail).

Hence, when $X = x$ ($x \neq 1$), we have $f(x) = 2 \cdot (\frac{1}{2})^x = 2^{1-x}$ (The probability of one permutation of tails and heads in x flips is $(\frac{1}{2})^x$)

So the pmf f is:

$$f(x) = \begin{cases} 0 & \text{if } x = 0, 1 \\ 2^{1-x} & \text{if } x \neq 0, 1 \end{cases} \quad (2)$$

(b)

Suppose the mgf to be $M_X(t)$, then:

$$\begin{aligned} M_X(t) &= E[e^{tX}] \\ &= \sum_{x=2}^{\infty} e^{xt} f(x) \\ &= \sum_{x=2}^{\infty} e^{xt} 2^{1-x} \\ &= 2 \sum_{x=2}^{\infty} \left(\frac{e^t}{2}\right)^x \end{aligned}$$

The infinite geometric series converge if and only if $|\frac{e^t}{2}| = \frac{e^t}{2} < 1$, i.e. $t < \ln 2$.

And the sum (if exists) (First term is $\frac{e^{2t}}{4}$, common ratio $r = \frac{e^t}{2}$) is obtained by:

$$M_X(t) = 2 \frac{\frac{e^{2t}}{4}}{1 - \frac{e^t}{2}} = \frac{e^{2t}}{2 - e^t} \quad (\text{where } t < \ln 2)$$

(c)

By taking 1st and 2nd derivative, we obtained:

$$\begin{aligned} M'_X(t) &= -\frac{(e^t - 4)e^{2t}}{(e^t - 2)^2} \\ M''_X(t) &= -\frac{e^{2t}(e^{2t} - 6e^t + 16)}{(e^t - 2)^3} \end{aligned}$$

Let the mean be μ and the variance be σ^2 , then:

$$\begin{aligned}\mu &= M'_X(0) = 3 \\ \sigma^2 &= M''_X(0) - (M'_X(0))^2 = 11 - 3^2 = 2\end{aligned}$$

(d)

We can use the results in (a) directly:

$$\begin{aligned}P(X \leq 3) &= P(X = 2) + P(X = 3) = \frac{1}{2} + \frac{1}{4} = \frac{3}{4} \\ P(X \geq 5) &= 1 - P(X \leq 4) = 1 - P(X = 2) - P(X = 3) - P(X = 4) = \frac{1}{8} \\ P(X = 3) &= 2^{1-3} = \frac{1}{4}\end{aligned}$$

2.4-18

Let X be the number of required test, n be the number of individuals, p be the probability of a person having the disease.

The test result of the common pool is negative if and only if none of the individuals have this kind of disease, and we only need $X = 1$ test, also:

$$P(X = 1) = (1 - p)^n$$

The test result of the common pool is positive if and only if at least one individual has this disease, and we need $X = n + 1$ test, also:

$$P(X = n + 1) = 1 - (1 - p)^n$$

Hence, the expected value of X (given $p = 0.05, n = 5$) is:

$$E[X] = 1 \cdot (1 - p)^n + (n + 1) \cdot (1 - (1 - p)^n) \approx 2.13110$$

The expected value of the number of tests needed is (approximately) 2.13110.

2.6-9**(b)**

For the convenience of calculation, I do (b) first.

Define the random variable X be the number of requests for newspaper per day, and f be the p.m.f. By the formula for Poisson distribution:

$$P(X \leq n) = \sum_{x=0}^n f(x) = \sum_{x=0}^n \frac{\lambda^x e^{-\lambda}}{x!}, \quad P(X > n) = 1 - P(x \leq n)$$

By some calculation:

$$\begin{aligned} P(X = 0) &= e^{-3} \approx 0.04979 \\ P(X = 1) &= 3e^{-3} \approx 0.14936 \\ P(X = 2) &= \frac{3^2 e^{-3}}{2!} \approx 0.22404 \\ P(X = 3) &= \frac{3^3 e^{-3}}{3!} \approx 0.22404 \\ P(X = 4) &= \frac{3^4 e^{-3}}{4!} \approx 0.16803 \\ P(X = 5) &= \frac{3^5 e^{-3}}{5!} \approx 0.10081 \\ P(X = 6) &= \frac{3^6 e^{-3}}{6!} \approx 0.05041 \end{aligned}$$

We found:

$$P(X > 5) = 0.08393 > 0.05, \quad P(X > 6) = 0.03353 < 0.05$$

Thus, the minimum number the manager should order is 6.

(a)

Let X be the number of request per day, and we are given that X follows a Poisson distribution with a mean of 3.

Since the store only stocks $n = 4$ newspaper. Let Y be the number of newspaper actually sold, then $Y = \min(X, 4)$.

We then can calculate the expected value of Y :

$$\begin{aligned} E[Y] &= \sum_{x=0}^3 x \cdot P(X = x) + \sum_{x=4}^{\infty} 4 \cdot P(X = x) \\ &= 0 \cdot P(X = 0) + P(X = 1) + 2 \cdot P(X = 2) + 3 \cdot P(X = 3) + 4 \cdot P(X \geq 4) \\ &= 0 \cdot P(X = 0) + P(X = 1) + 2 \cdot P(X = 2) + 3 \cdot P(X = 3) + 4 \cdot \left(1 - \sum_{x=0}^3 P(X = x)\right) \\ &= 4 - 4 \cdot P(X = 0) - 3 \cdot P(X = 1) - 2 \cdot P(X = 2) - P(X = 3) \\ &= 4 - 4 \cdot 0.04979 - 3 \cdot 0.14936 - 2 \cdot 0.22404 - 0.22404 \\ &= 2.68064 \end{aligned}$$

So the expected of the number sold is 2.68064.